

1. Client Submission Interface

Streamlit Ingestion UI

- Contributor Sinhala input
- Informed consent & safety tags
- Real-time character telemetry



Pre-Commit Quality Gate

- Text Normalization (NFC)
- Script & length bounds (50-100k)
- Deduplication checks:
 - Exact SHA-256 vs hash index
 - Jaccard similarity vs pending
- AI-generated text suspicion check



Metadata Packaging

- ID: SLS-YYYYMMDD-HEX
- UTC timestamp & SHA-256
- Session hash & safety flags
- Encoded to isolated .jsonl file

Local Resilient Queue (Offline)

Local disk buffer with auto-replay on reconnect

2. Staged File Queue

Conflict-Free Ingestion

- Zero concurrent write conflicts
- Decoupled from central dataset
- Low latency client commit (<1s)

Remote pending/ Directory

Isolated micro-commit per submission:

entry_20260901_a1f4_01.jsonl

Atomic staged record (Submission 1)

entry_20260901_b7c2_02.jsonl

Atomic staged record (Submission 2)

entry_20260901_e9d8_03.jsonl

Atomic staged record (Submission 3)

entry_20260901_fa33_04.jsonl

Atomic staged record (Submission 4)

Queue Properties

- Hosted on Hugging Face Hub
- Append-only staging area
- Independent atomic files
- Immune to parallel push failures
- Persisted until automated merge

3. Automated Merge Workflow

Scheduled Batch Ingestion

- Triggered every 5 min (cron)
- Batch downloads all pending/ files
- Dynamic schema alignment & union



Optimised Incremental Merge

- Loads train-append.parquet only
- Bypasses 10.9M+ row core dataset
- Lightweight execution (<15s)
- Low RAM & compute footprint



Index & Statistics Sync

- Appends SHA-256 to hashes.txt
- Mathematical updates to:
 - dataset_stats.json
 - merge_stats.json (telemetry)

Atomic Hub Commit & Purge

Single atomic commit operation:

1. Write Parquet, stats & hashes
2. Delete merged pending/ files

-> Eliminates race conditions & duplicate data

Real-time Telemetry & Dataset Statistics Feedback Loop (dataset_stats.json)